HYUNJE YANG

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Short version. Full CV available at hyunjeyang.com

EDUCATION

The University of Texas at Austin (UT Austin) — **Austin, TX, USA**
Ph.D. Candidate in Civil, Architectural and Environmental Engineering — *Aug. 2023 – Present*
- *UT Austin Engineering Fellowship, Cockrell School of Engineering and UT Austin Graduate School*
- *Teaching Assistant, Probability and Statistics for Civil Engineers (Spring 2025, 2026)*

Seoul National University (SNU) — **Seoul, South Korea**
M.S. in Forest Environmental Science — *Mar. 2016 – Feb. 2018*
- *Thesis: Formula for Calculating Mean Velocity in Mountain Streams Using the Salt Dilution Method*
- *Teaching Assistant, Forest Geological Information (Spring 2016)*

B.S. in Forest Environmental Science — *Mar. 2012 – Feb. 2016*
- *National Scholarship Award for 7 semesters, Ministry of Science and ICT*

RESEARCH INTERESTS

- Transferable AI Modeling
- Forecast Uncertainty and Early Warning
- Compound Coastal Flooding
- Hydrology of Ungauged Systems

RESEARCH EXPERIENCE

Oak Ridge National Laboratory (ORNL) — **Oak Ridge, TN, USA**
Graduate Research Intern — *May 2026 – Jul. 2026*
Research Collaborator · Austin, TX, USA (Remote) — *Jul. 2026 – Present*
- Geometry-aware neural operators to scale storm-surge prediction from local coastal domains toward regional-to-global coverage on unstructured meshes
- Investigating coastal wetland resistance and recovery using global tropical-cyclone records and satellite observations
- Developing a diffusion-based surrogate for reconstructing historical storm-surge dynamics

The University of Texas at Austin (UT Austin) — **Austin, TX, USA**
Ph.D. Student, Civil, Architectural and Environmental Engineering — *Aug. 2023 – Present*
- Mask-Aware Multimodal Deep Operator Network for Spatially Contiguous Mapping of Soil Water Retention Curves
- Texas-Scale Assessment of Multi-Driver Compoundedness in Coastal Flooding
- Physics-Informed Machine Learning for Compound Flooding Prediction
- Integration of Machine Learning and Static Flood Models for Coastal Inundation Simulation
- Reconstruction of Missing Data for Hurricane-Damaged Buildings using XGBoost

National Institute of Forest Science (NIFoS) — **Seoul, South Korea**
Researcher, Forest Environment and Conservation Department — *Mar. 2018 – Apr. 2023*
[Government R&D Projects]
- Establishment of Big Data and Integrated Utilization System for Flash Floods in Forest Watershed
- Development of Forest Water Management Technology to Reduce Stream Depletion and Nonpoint Source Pollution
- Quantification and Improvement of Forest Water Yield for Sustainable Supply of Freshwater
- Long-term Monitoring of Flow Characteristics in Different Forest Stands and Locations

Seoul National University (SNU) — **Seoul, South Korea**
Researcher, Department of Forest Environmental Science — *Jun. 2013 – Feb. 2018*
- Hydraulic Relation of Discharge and Velocity in Mountain Streams Using Salt Dilution Method
- Monitoring of Floods in Mountainous Areas and Critical Runoffs in Urban Areas

PUBLICATIONS

[Under Review]

- **Yang, H.**, Lee, J.W., Lee, W. "Where, When, and How Do Physics-Informed Deep Learning Models Improve Hurricane-Induced Compound Flooding Predictions?" *ESS Open Archive*, 2026. https://doi.org/10.22541/essoar.15005345/v1

[In Preparation]

- **Yang, H.**, Li, Z., Lee, J.W., Kim, T., Tran, V.N., Lu, D., Dias, P.A., Painter, S.L., Le, P.V.V. "Geometry-Aware Neural Operators for Transferable Storm Surge Prediction" *In preparation.*
- Li, Z., **Yang, H.**, Dias, P.A., Faxon, A., McCarthy, M., Sulman, B., Painter, S.L., Tran, V.N., Lu, D., Le, P.V.V. "Divergent Controls on Coastal Wetland Resistance and Recovery to Tropical Cyclones" *In preparation.*
- Le, P.V.V., Dias, P.A., **Yang, H.**, Li, Z., Tran, V.N., Lu, D., Faxon, A., McCarthy, M., Sulman, B., Painter, S.L. "Learning Transferable Storm Surge Dynamics from Synthetic Hurricanes Using Diffusion Models" *In preparation.*
- Chowdhury, M.E., **Yang, H.** "Analysis of Flood-Driver Transition across 14 Historical Storms in the Houston–Galveston Region" *In preparation.*
- **Yang, H.***, Kim, T.*, Lee, J.W., Ivanov, V. "HT-PTF: Landscape-Informed Operator Learning for Spatially Contiguous Mapping of Soil Water Retention Curves" *In preparation.* (*Both authors contributed equally to this work)

[Journals]

- **Yang, H.**, Lee, J.W., Santos Cruz, A.U. "Machine Learning-Enhanced Static Flood Models for High-Resolution Peak Storm Surge Inundation Mapping in Southeast Texas, USA" *Journal of Hydrology: Regional Studies*, 2026, 63: p. 103056. https://doi.org/10.1016/j.ejrh.2025.103056
- **Yang, H.**, Lee, J.W., Klepac, S., Santos Cruz, A.U., Subgranon, A., Jiao, J. "Reconstructing Missing Data of Damaged Buildings From Post-Hurricane Reconnaissance Data Using XGBoost" *Frontiers in Built Environment*, 2024, 10: p. 1444001. https://doi.org/10.3389/fbuil.2024.1444001
- **Yang, H.**, Lim, H., Moon, H., Li, Q., Nam, S., Choi, B., and Choi, H.T. "Identifying the Minimum Number of Flood Events for Reasonable Flood Peak Prediction of Ungauged Forested Catchments in South Korea" *Forests*, 2023, 14(6): p. 1131. https://doi.org/10.3390/f14061131
- Nam, S., **Yang, H.**, Lim, H., Kim, J., Li, Q., Moon, H., and Choi, H.T. "Short-Term Effects of Forest Fire on Water Quality Along a Headwater Stream in the Immediate Post-Fire Period" *Water*, 2023, 15(1): p. 131. https://doi.org/10.3390/w15010131
- **Yang, H.**, Lim, H., Moon, H., Li, Q., Nam, S., Kim, J., and Choi, H.T. "Simple Optimal Sampling Algorithm to Strengthen Digital Soil Mapping Using the Spatial Distribution of Machine Learning Predictive Uncertainty: A Case Study for Field Capacity Prediction" *Land*, 2022, 11(11): p. 2098. https://doi.org/10.3390/land11112098
- **Yang, H.**, Yoo, H., Lim, H., Kim, J., and Choi, H.T. "Impacts of Soil Properties, Topography, and Environmental Features on Soil Water Holding Capacities (SWHCs) and Their Interrelationship" *Land*, 2021, 10(12): p. 1290. https://doi.org/10.3390/land10121290
- Lim, H., **Yang, H.**, et al. "Development of Pedo-Transfer Functions for the Saturated Hydraulic Conductivity of Forest Soil in South Korea Considering Forest Stand and Site Characteristics" *Water*, 2020, 12(8): p. 2217. https://doi.org/10.3390/w12082217
- **Yang, H.**, Choi, H.T., and Lim, H. "Effects of Forest Thinning on the Long-Term Runoff Changes of Coniferous Forest Plantation" *Water*, 2019, 11(11): p. 2301. https://doi.org/10.3390/w11112301
- **Yang, H.**, Choi, H.T., and Lim, H. "Applicability Assessment of Estimation Methods for Baseflow Recession Constants in Small Forest Catchments" *Water*, 2018, 10(8): p. 1074. https://doi.org/10.3390/w10081074
- **Yang, H.**, Lee, S., and Im, S. "Hydraulic Relation of Discharge and Velocity in Small, Steep Mountain Streams Using Salt Dilution Method" *Journal of Korean Society of Forest Science*, 2018, 107(2): p. 158-165. https://doi.org/10.14578/jkfs.2018.107.2.158
- **Yang, H.**, Im, S., Lee, Q., Eu, S., and Lee, E. "Flow Measurement of Mountain Streams using Salt Dilution Method" *Journal of the Korean Society of Forest Engineering*, 2016, 14(1): p. 1-6. https://www.earticle.net/Article/A276485

CONFERENCE PRESENTATIONS

[Oral Presentations]

- **Yang, H.**, Li, Z., Lee, J.W., Kim, T., Le, P.V.V. "Multi-GPU Training of Geometry-Aware Neural Operators for Cross-Regional Storm Surge Prediction" TACC Symposium for Texas Researchers 26, Texas, USA, Sep. 21-22, 2026.
- **Yang, H.**, Lee, J.W., Santos Cruz, A.U., Subgranon, A., Klepac, S.P., Jiao, J. "Can Hurricane, Surge, and Damage Data Assist in Reconstructing Structural Features?" 10th Young Coastal Scientists and Engineers Conference - Americas, Quebec, Canada, Jun. 4-6, 2024.
- **Yang, H.**, Lee, J.W. "Towards Resilient Texas: Rapid Storm Surge Prediction Using Machine Learning" Planet Texas 2050 Symposium, Texas, USA, Feb. 27-29, 2024.

- **Yang, H.**, Choi, H.T., and Lim, H. "Estimating Flood Peaks in Ungauged Forested Catchments" Joint Symposium on Forest Science, Seoul, Korea, Feb. 15-16, 2023.

[Poster Presentations]

- **Yang, H.**, Lee, J.W., Lee, W. "Development of a Physics-Informed Deep Learning Model for Hurricane-Induced Compound Flooding Prediction" HydroML 2026, Texas, USA, May 19-21, 2026.
- **Yang, H.**, Lee, J.W., Lee, W. "Spatiotemporally Heterogeneous Improvements in Compound Flooding Prediction through Physics-Constrained Learning and Their Implications for Coastal Risk" Project SHIELD Workshop, Texas, USA, May 7, 2026.
- Kim, M., **Yang, H.**, Chowdhury, M.E., Lee, J.W. "Compound Flooding and Power Grid Resilience in Puerto Rico: Evaluating Extreme Forcing Duration using Mixed Hurricane Irma and Maria Scenarios" 2026 Energy Research Poster Competition, Texas, USA, Apr. 6-7, 2026.
- **Yang, H.**, Lee, J.W., Santos Cruz, A.U. "A Hybrid AI-Driven and Static Flood Modeling for Real-Time Peak Storm Surge Inundation Mapping" American Geophysical Union (AGU) Fall Meeting 2025, Louisiana, USA, Dec. 15-19, 2025.
- **Yang, H.**, Lee, J.W., Santos Cruz, A.U. "A Hybrid Approach Integrating Machine Learning and Static Flood Models for Efficient Real-Time Peak Storm Surge Inundation Mapping" Artificial Intelligence and Digital Twins for Earth Systems, Texas, USA, Sep. 22-24, 2025.
- **Yang, H.**, Lee, J.W. "Storm Surge Hazard Estimation for Texas Coastal Areas Using Machine Learning" 2024 CAEE Graduate Symposium, Texas, USA, Feb. 23, 2024.

PEER-REVIEW SERVICE

- Journal of Hydrology
- Geomatics, Natural Hazards and Risk

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, ArcGIS, QGIS
- **Modeling:** Geometry-aware neural operators, hydrodynamic models (SFINCS, ADCIRC), physics-constrained deep learning, diffusion models
- **High-Performance Computing:** GPU-accelerated model training and inference on HPC systems; multi-GPU and distributed training workflows
- **Experimental and Field:** Installation of hydrological instruments and data collection via datalogger, water level gauge, sap flow sensor, and automatic water sampler; salt-dilution stream discharge measurement; forest soil sampling and hydraulic property analysis

REFERENCES

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